

Internship portfolio report

1. Description of the company and role

1.1 Description of the company:

Lely (officially Lely Industries N.V.) is a Dutch, family-owned agricultural technology company founded in 1948 by brothers Cornelis and Arij van der Lely in Maassluis, the Netherlands. It has evolved into a global leader in automated solutions for dairy farming, focusing on innovation to make farm work easier, more efficient and more sustainable. The company operates in more than 50 countries and has around 4000 employees worldwide.

1.2 Vision:

Lely's vision is centered on creating a sustainable, profitable and enjoyable future in farming. It aims to lead agricultural innovation by developing technologies that improve animal welfare, environmental impact, dairy quality and farmer prosperity.

1.3 History:

- 1948: Founded by Cornelis and Arij van der Lely with the goal of making life easier for farmers through innovation.
- Initially developed simple agricultural tools; over time it expanded into sophisticated automated systems.
- 1992: Debut of the Lely Astronaut, one of the first commercial robotic milking systems, which revolutionized dairy farming.
- Lely has since grown its range to include feeding, barn cleaning and herd management automation.
- Today it holds 1,500+ active patents and maintains a massive global presence with dedicated Lely Centres supporting farmers locally.

1.4 Products:

Lely develops and manufactures innovative automated and digital solutions for dairy farms, including:

- Milking robots (e.g., Lely Astronaut A5 series) enabling cows to be milked on demand.
- Feeding systems (e.g., Vector, Juno Max) optimising feeding routines and nutrition.
- Manure and barn handling systems (e.g., Discovery Collector C2).
- Monitoring and management systems (digital tools for herd and barn oversight).
- Services and advice on farm planning and efficient operation.

1.5 Market and target group:

- **Market:** Global agricultural machinery and technology market with emphasis on dairy automation. Lely's solutions are used in more than 50 countries across Europe, the Americas, Asia and beyond.

- **Target Group:** Primarily dairy farmers and livestock producers seeking to increase efficiency, productivity, and sustainability through automation and data-driven farm management. This includes small family farms and larger commercial dairy operations. Lely also serves through a network of Lely Centers, offering local sales, implementation, and after-sales support.

1.6 My project and role in the company:

My specific role in the company and the project I will be mainly working on is called the Beaver project, formerly known as Milk Interval Optimization (MIO). I will work within the Data & AI department in the company, which has about 40 people in it. I will be working in the Milk Settings Optimization (MSO) team which has 4 other people besides me, and I will be the first intern to be filling out this position for this period. The MSO team has a few other projects that they have going on, but like I stated I will be mostly working on the Beaver project.

1.7 Beaver project:

The beaver project was created to support the Horizon application, which is an application farmers can download on their phones that supports them in using the Lely Astronaut machines. Currently the farmers must manually configure when the dairy cows are to be milked on an outdated and confusing page, that is used for milk access. If farmers continue to manually control when the cows are milked there is bound to be variation and performance loss. The solution that the Beaver project brings is a fully automated milk access timing at cow level, the system determines when each cow should be milked using prediction models, udder refill patterns, behaviour data, and robot load optimisation. And overall, it is designed to wholly improve the milking process for farmers and provide them with extra profits and milk yield. I will be working on improving this work, working on improving their models and predictions, as well as explore the data and uncover new patterns and insights related.

Organizational chart:

Company organizational chart for the AI team

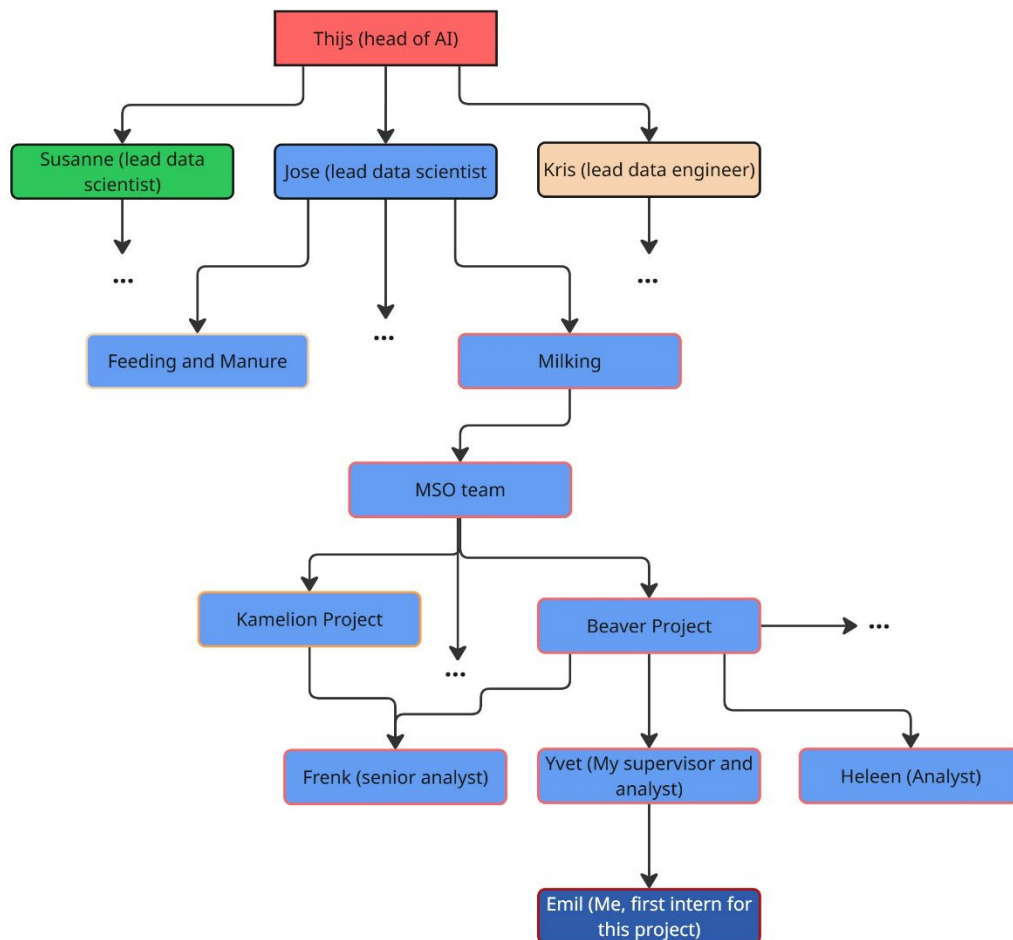


Fig.1. An organizational chart of the Data and AI team, making one for the whole company is outside of the scope and it also highlights my role in the company.

1.8 Work culture and environment:

The way I would describe the culture and environment at the workplace is quite relaxed and open, at least for the AI team which is the only sector of the company I can speak about. Everyone was very welcoming and there were colleagues of all backgrounds, nationalities and ages, and there were very few times in which I felt excluded or uncomfortable.

The work culture and the way things work here, the systems that are in place I would describe as being quite free flowing. Everyone has their sub-team and tasks they need to do, and everyone works hard to complete their tasks and help the team grow in every way they can, meaning it is a productive work environment but without making it feel too robotic and work heavy at the same time.

There are regular monthly events and meetings with different subsets of the teams where we all gather to talk about certain subjects, catch up, brainstorm, etc. In my team specifically, which consisted of 5 people, we had weekly sprint meetings, bi-weekly team meetings, project sync calls, weekly standups with everyone and I also had regular one-on-ones with my supervisor. And whenever I tried to setup meetings with other people from the AI department, they were always open to it and even made time for it sometimes. In the weekly sprint planning every task from each team member is iterated over for the coming week with completion statuses for each task.

2. Ethical and Legal considerations

2.1 Databricks:

Over the course of my internship period, I worked for most of the time within Data Bricks. Data Bricks is a cloud-based platform for data analytics and artificial intelligence. Databricks developed the 'data lakehouse' architecture, which combines elements of data warehouses and data lakes for managing structured and unstructured data.

Lely leverages Databricks to revolutionize dairy farming by transforming vast amounts of data from robotic milking systems into actionable insights. Using a "data product" approach with the Databricks' lakehouse platform, Lely reduces data implementation time by up to 90%, optimizing cow health, farm efficiency, and data-driven innovation in agriculture. Lely, a leader in robotic milking, treats data as a product (data + metadata), which enables them to quickly share insights across the organization, accelerating the development of new applications.

All Lely employees are trying to be compliant with certain things, and in terms of compliance there is an agreement with Data Bricks about data confidentiality and the security of the data. Data Bricks is also linked to a storage account and that account and all the resources connected with Data Bricks is being handled in terms of security by azure on the cloud level. Data Bricks also keeps the lineage of data and the flow of the data, to see where it starts and ends.

Terraform is also used for managing and deploying resources, and most things of that nature are managed there. Terraform an open-source infrastructure as code (IaC) tool that allows you to define, provision, and manage cloud and on-premise infrastructure using a declarative configuration language (HCL). With Terraform, it is used to granularly define permissions on the catalogue of Data Bricks and all of its data bases and tables.

2.2 Transparency & Explainability in Data:

For Lely, in the general sense, transparency and explainability is important. Not specifically in a very data centric way but the way the data appears for farmers, where it comes from, and how it is used. At Lely, using the large amounts of data, for a specific farm for example there is always data being generated from it and data stored from that farm. So, we are able to see what kind of metrics it is producing and what metrics/farm strategies the farmers use and if

an anomaly comes up or something worth fixing/exploring, we the programmers can send optional 'advices' to the farmers to follow, for the most optimal outcomes. So, it is important to have the data and the solutions we provide be completely transparent and explainable in order to convince the farmers to use those different strategies and provided solutions, especially for those stubborn farmers who aren't fond of new technologies.

2.3 Data Governance:

The core goal of data governance at Lely is to ensure data is managed in a way that generates value, not for its own sake, but tied directly to any business goals, process improvement, and risk reduction. To make all this more easily manageable, all of the Lely data is divided into 13 data domains (e.g. customer supplier, milking, feeding). Within those domains, data is further split into data products, which is any set of data actively used in the organisation. The purpose of this division is for accountability, meaning there will always be a clear owner for data quality and management. Without it nobody could answer who is responsible for data quality or who to go to when issues arise.

The key roles for Data Governance include:

- **Data Domain Owner** - sits within the business itself, closest to the actual processes and the data those processes generate, responsible for maintaining data quality day-to-day
- **Data Product Owner** - operates at the tactical level, accountable for decisions about a specific data product
- **Business data steward** - sits within the business itself, closest to the actual processes and the data those processes generate, responsible for maintaining data quality day-to-day
- **Technical data steward** - the application expert, responsible for the storage, lineage, and technical management of data within a specific tool or platform (e.g. Databricks). One person can hold multiple roles simultaneously.

2.4 Compliance & Regulations:

Within Lely there is an information classification policy that puts all kinds of data and information into 4 different categories: highly confidential, confidential, internal, and public, with documented definitions for each one. But anything beyond that and compliance becomes less structured. For external regulation GDPR is the primary choice for the company to follow, and there is also a data privacy officer role being introduced into the company, but there is no formal or standardised procedure ensuring organisation-wide compliance. People working with personal data are expected to consult either their domain owner or the privacy officer, but this does not always happen in practice. The EU AI Act is on the radar and being discussed internally, but much like with the case of GDPR, it does not have a formal compliance process in place yet at the company. An external consultant is currently drafting AI principles for the organization, though these are not yet finished or published.

2.5 Maturity Level & Ethical Considerations:

After assessing the companies legal and ethical considerations, it seems Lely is still at a very early, initial phase of data governance maturity. Topics like fairness and bias in data and algorithms, such unrepresentative training data or historical discrimination are not currently being actively tested or monitored. In the future, as the company grows larger, the expectation is that as the governance structure matures and the domain/product owner/steward roles become more established at the company, these responsibilities will naturally be picked up at the appropriate levels. But for now, the focus is to get the foundational structure in place first.

3. Internship Task: Explanation and Approach

In dairy farming, the dry-off period refers to the phase in a dairy cow's reproductive cycle during which milking is deliberately ceased, allowing the cow's mammary gland to rest and regenerate before the next lactation. This transition period, typically lasting around 6 to 8 weeks prior to calving, is critical for udder health, milk production in the subsequent lactation, and overall cow welfare. Effective dry-off management, including the timing of cessation and nutritional adjustments all play a significant role in determining the productivity and health outcomes of the herd. As dairy operations scale and data availability grows, accurately predicting and managing milk production traits around the dry-off period has become an increasingly important area of research. What is typically done in end lactation stage of the cow that the milk yield is gently lowered during the end lactation phase to have a nice transition to the dry period and that yield is lowered by restricting milk access

My internship focuses a lot on the dry-off periods of the cow, and in general I am confined to the end lactation stage of the cow. My task was to gain insights and findings into this period of the cow using the wide and vast datasets collected from the thousands of farms that use Lely robots for milking, and relate these findings to outcome variables such as the milk yield of the cows, or cow health and give strategies about moving forward with the dry-off process.

Relating all this domain knowledge to my own skills and background, we decided the way to gain the most insights was to use clustering. Specifically, I chose K-Means as the algorithm I was going to use to cluster different settings used by the farms in the end stage of lactation. Each farm and group of cows had a different combination of these settings, so the goal was to use those groups of settings that each farmer has set themselves and perform a cluster analysis with them in mind. Then I would get back the most used combination of settings and strategies that each farmer uses and relate it to certain effects like daily milk yield from the animals during the end-lactation. And with that I could get cluster profiles from gathering the combination of those settings and gaining an insight into why a farm would use a specific kind of strategy for their cows.

4. Integration of research and theory

The use of Machine Learning (ML) in the wide field of Agri-tech is nothing new for the industry, for both agricultural companies and in the field of Data Science and AI. Focusing more specifically on the application of machine learning to dairy farming and Automated Milking Systems (AMS). All of it spans a range of objectives, from disease detection and herd characterisation to milk yield prediction and farm-level classification. This review synthesises findings from relevant studies, examining the methods employed, results achieved, and gaps that remain in the literature.

A broad review made by Ozella et al (2023) of 60 other studies that used ML algorithms and applications in AMS revealed that there was an imbalance in research focus in the field. The existing work focused mostly on disease detection, the research lacking topic being for milk production, the one I will focus on. This highlights a clear gap in research and a lack of attention towards milk production in the field of ML and dairy farming. This is surprising given that AMS systems continuously generate rich production data that could meaningfully support farm management decisions.

Many of the studies that were looked at explored AMS data for herd characterization, and to monitor cows' behaviour to uncover and develop unique patterns overtime. Ozella et al (2023) references the work done by Adamczyk et al., in which the physical activity of dairy cows milked in the voluntary milking system recorded by neck-mounted tags was analysed using ML methods. What they did was perform a cluster analysis, classifying the physical activity of cows using Ward's hierarchical agglomeration method and Kohonen self-organising maps. Physical activity during individual months showed small variability. However, over the individual months, the cluster obtained by Ward's method highlighted different groups depending on daytime light length, temperature, and relative humidity. In that study, Kohonen networks were used only for the verification of Ward's method and to test the similarity between clusters obtained with both methods.

The same review also highlights K-means clustering applied to herd segmentation using five parameters, daily milkings, parity, activity, milking regularity, and body weight, demonstrating AMS data's suitability for automated, management-relevant grouping.

At the farm level, Žgajnar (2024) applied a combined hierarchical and K-means approach to classify Croatian dairy farms into representative archetypes for policy analysis. Their results showed that variable selection significantly affected cluster interpretability: a reduced two-variable model produced more logically ordered clusters than the full four-variable version, ranking farms cleanly by herd size and milk output.

Atıl and Akıllı (2016) directly compared K-means and artificial neural networks (ANNs) for classifying 10,000 dairy cattle by milk production traits. The ANN used a multilayer perceptron trained with three backpropagation algorithms, of which Bayesian regularisation performed best. The ANN achieved a peak R^2 of 97.8%, compared to 94.9% for K-means, though performance varied across datasets, dropping to 72.3% on one, indicating sensitivity

to data characteristics. Overall, the ANN demonstrated a better fit, suggesting neural networks are a viable alternative to traditional clustering for animal classification.

Ji et al. (2022) developed an XGBoost-based forecasting framework using five years of data from 80 cows in a robotic dairy farm, predicting daily milk yield, milking frequency, protein content, and fat content up to 28 days ahead. Validated across 42 cross-validation folds, results were strong: protein content reached ~98% accuracy, fat content ~95%, and daily milk yield maintained a mean above 90%. Feature importance analysis showed cow-level features were considerably more predictive than climate features, and cow age outweighed lactation stage, attributed to the shortened productive lifespan of modern cows creating pronounced age-related production differences. A key limitation was systematic bias at extremes, with the model overestimating low values and underestimating high ones, a common pattern in large-dataset regression models.

4.1 Synthesis

Across the literature, K-means combined with Ward's method emerges as a consistent and versatile tool for both cow- and farm-level segmentation. ANN approaches outperform K-means in direct comparisons, though not uniformly. For production prediction, XGBoost shows strong performance when cow-level features are prioritised. The central gap identified by Ozella et al. (2023) and reflected across all reviewed studies is the underrepresentation of production-focused ML research relative to disease detection. Future work could benefit from integrating clustering and prediction within a unified framework, using herd segmentation outputs as contextual features for individual-level forecasting.

5. Methodology

5.1 Exploratory Data Analysis

Before I started doing the cluster analysis, which is what would lead to the result and final deliverable of my task I had to do some data analysis, asking questions that were in and around my main task to first gain some insights and context on my task, and to gain a better understanding on the kind of data I was working on so I could move on to the main part of my work. This step of the process was also used by me to the domain heavy knowledge, for the purpose of getting a real grasp on the data and how to use it before I got into clustering. The first step is to always understand the data.

The main language used was SQL for all pre and post analysis, all done within Databricks. The methodology for the analyses was to first form questions around the main topic that I tried to answer with the data, that could later help with the main task.

It all stemmed from a big query given to me by the team that had the features/settings I used for the clustering all calculated there. After exploring the data enough and preparing the main query to be used for the main task, I saved it as a table to be used more accessibly later on and then I started on the main task.

5.2 ML tools

The cluster analysis was conducted using four key settings that farmers configure for their cows during the end lactation phase that came from the milk access table (MAS). These settings were selected as input variables as they directly reflect the dry-off strategy applied by each farm:

- **Days until dry-off:** the point at which a cow transitions from mid to end lactation in the milk access table
- **Minimum number of milkings in end lactation:** the lower bound of milking frequency permitted during the end lactation phase
- **Maximum number of milkings in end lactation:** the upper bound of milking frequency permitted during the end lactation phase
- **Optimum expected yield in end lactation:** the target milk yield set for a cow during the end lactation phase

Then to deduce the number of clusters to use I tried the elbow method, which measures the squared distance between data points and their assigned cluster centroids. The most common number of clusters I kept getting was between $K = 4-5$. Whenever I tried using more clusters, the latter end of them, mostly the clusters 6 and upwards would always overfit and then it would underfit with too few clusters.

To be able to graph all the clusters as points on a scatter plot, and due to the nature of this case being more than 3 dimensions (4 settings were involved, more than 3) I had to conduct a Principal Component Analysis (PCA) in order to project the results on a 2D plain. Then I combined the settings into Principal Components (PC), going by how much they affect the results, so the most influential settings get grouped into PC1 and the secondary ones into PC2.

5.3 Quality and Reliability

Throughout the process of carrying out my internship tasks and project, while I was writing code logic, exploring new ideas and insights that revolved the main project, any model that I decided to develop, to ensure real validity and quality to the solutions I provided or methods I chose to work with I always made sure to check my work over with my supervisor and the rest of my team to make sure everything is up to standard and that I wasn't receiving false information that would hurt the analyses.

Since I came into this with little to no knowledge regarding the dairy and farming industry, making sure to check everything over with them was of utmost importance to ensure no metrics were subject to hallucinating values and that the results I was getting were realistic and always made sense.

In addition to consulting with my team and colleagues I consistently performed data validation checks to ensure the correctness of the datasets used.

5.4 Adding a new feature

After discussing and analysing the previous results I decided on adding a new layer to my analysis. That came in the form of making a new feature or setting for the clustering analysis. This was done to try and improve the clustering that was done, the main basis of this was to capture farm-level grouping behaviour related to late-pregnancy cows. Since this whole project was focusing on late-stage cows, adding this extra layer would allow an enrichment to the analysis into late-stage lactating cows. The goal of this feature is to incorporate biological and management-based context into the clustering, beyond purely configuration-based MAS parameters. This flag is created to detect whether the farm has artificially created an additional segment in the milk access settings table. What happens is that instead of the normal 3 segments (start, mid and end), they move the cows halfway to another group and therefore create 4 segments (start, mid, mid in the new group and end).

While the current variables describe the system configuration and touch up on very important features used for the different kinds of farming strategies that are actively used, they do not capture how farms organize cows into groups, whether farms use late-pregnancy grouping strategies and how management aligns biological state with MAS rules. To address this gap this new feature was added.

The new setting was made with a combination of two separate metrics:

- `animal_days_pregnant_from` ≥ 200
- `animal_days_pregnant_to` ≥ 200

These features indicate whether the farm uses late-pregnancy rules, defined as the two metrics above. These groups correspond to cows approaching the dry-off phase, where milk access restrictions are typically most relevant. The ≥ 200 means that I filtered for cows that are above 200 days into the late-pregnancy rule usage.

The way I implemented it into the main SQL query used for the cluster analysis as a feature was by transforming it into a percentage value. It starts by checking the ≥ 200 rule for either case and assigning a 1 or a 0 depending on if there is any record of the rule's usage for any cow, and then that binary score is averaged out for the entire cluster, then $\times 100$ to get the percentage value out of it. And with that the new feature is obtained.

5.5 Extra considerations

5.5.1 Important flags

For the main SQL query used to make the cluster analysis some important considerations had to be kept in mind for the validity of the results. That comes in the form of flags that some of the tables I used include. Those are:

- **is_current_version flag:** This flag checks if the settings and metrics of the cow that is being queried and studied in the current moment are the valid ones to use and the active true settings being used by the farmers for the cows for the current moment.

This is important because Lely keeps a record of historic settings used by all the cows, as these animals go through many stages of lactation during pregnancy. And it is always important to make sure I am making use of the current settings, otherwise none of the results can be used.

- **group_deleted_flag:** In every farm there are different sets of groups for different stages of cows, each with their individual name and ID. And often, when one of those groups has finished that state and there aren't any other ones to replace them for that group, the group is deleted. So, it is also important, in this case, to only keep existing groups into the analysis.
- **active_flag:** This, much like the other two, is important to have enabled to make sure the analysis only takes in cows that are currently active and being milked regularly. There is also a common table expression (CTE) at the start to only filter for animals that have been active in the last 40 days, that also happen to be in the late stage of lactation.

5.5.2 Thresholding

Much like in the point above, there were even more filtering measures to consider for the analysis. Instead of checking for binary flags, here we check a threshold of values as an extra strategy to use for the analysis. After checking the max threshold for each of my features in the horizon application, the main application and dashboard that farmers use to monitor the animals, provided by Lely, I could gather what the max setting that can be used within the application. This must be done because a lot of times farmers somehow input settings that are very irregular and can't be possible. Whether that is a fault of the system or the farmer not knowing what they are inputting or how they are doing it, it is important to set a cut off limit for realistic values to have and use within the analysis. The cut off values include:

- **Days until dry-off ≤ 30**
- **Minimum number of milkings in end lactation ≤ 21.6**
- **Maximum number of milkings in end lactation ≤ 21.6**
- **Optimum expected yield in end lactation ≤ 20**

5.6 Using ANN's

Earlier in the document, in the literature review section, when I was reading those papers, I had seen that in one of them that some people had used ANN's for clustering dairy cows. I decided to try that approach as well but with unsupervised machine learning algorithms instead.

From what I read, when the ANN was being used it demonstrated a better fit, suggesting neural networks are a viable alternative to traditional clustering for animal classification. Though in their paper they used a multilayer perceptron with clear target labels, I didn't have that, I only worked with unlabelled data. Thus, I chose to use an autoencoder.

Since an autoencoder only needs the raw data itself as input, it makes it suitable for clustering tasks where the ground truth labels don't exist. The reason for using an autoencoder in this

scenario is so that the network can learn the structure of the animal settings and give a more meaningful representation of the data.

6. Results

6.1 First general cluster analysis

The final table obtained from the clustering is the following:

Cluster	Max milk hr	Min milk hr	Milk yield kg	Dry off days
0	7.89	10.64	9.06	33.48
1	17.13	26.10	13.02	15.93
2	7.88	10.32	8.55	14.46
3	9.64	12.78	11.07	15.37
4	46.04	56.60	10.77	6.28

Fig.2. General cluster analysis table revealing all of the base settings

And the distribution of farms for each cluster is the following:

Cluster	Nr of farms
0	5850
1	1186
2	12377
3	10258
4	180

Fig.3. Table showing the distributions of farms per cluster

As it can be seen, the bulk of the farms are within cluster number 2 and 3, which happen to be the most standard combination of settings farmers seem to use. Then the more irregular settings are more concentrated in the other clusters, notably in the last one, where there is only a very small number of farms located inside of it. Going back to the table, that last cluster has by far the most irregular settings if you compare them to the other clusters, meaning the farmers that are in those clusters have quite the irregular combination of settings and strategies for their animals.

The distribution of weights for PC1 and PC2 loadings are the following:

Feature	PC1	PC2
Max milk hr	0.65	0.06
Min milk hr	0.64	0.11
Milk yield kg	0.34	0.18
Dry off days	-0.17	0.97

Fig.4. Table PCA loadings and distributions

From the table, we can observe that PC1 has strong positive loadings for max and min milkings hrs, so when a farm that is high for PC1 it usually means they are quite loose with their milking hours having a larger window for milkings. If it's low, then it means the milking time window is quite tight.

For PC2, it has positive loadings for milk yield and dryoff days. So, when a farm has high PC2 it means the milk yield is high and the dryoff window is long, and then the opposite of that if the PC2 values are low.

Here is the projected PCs on a scatterplot. Each point represents a farm from the dataset:

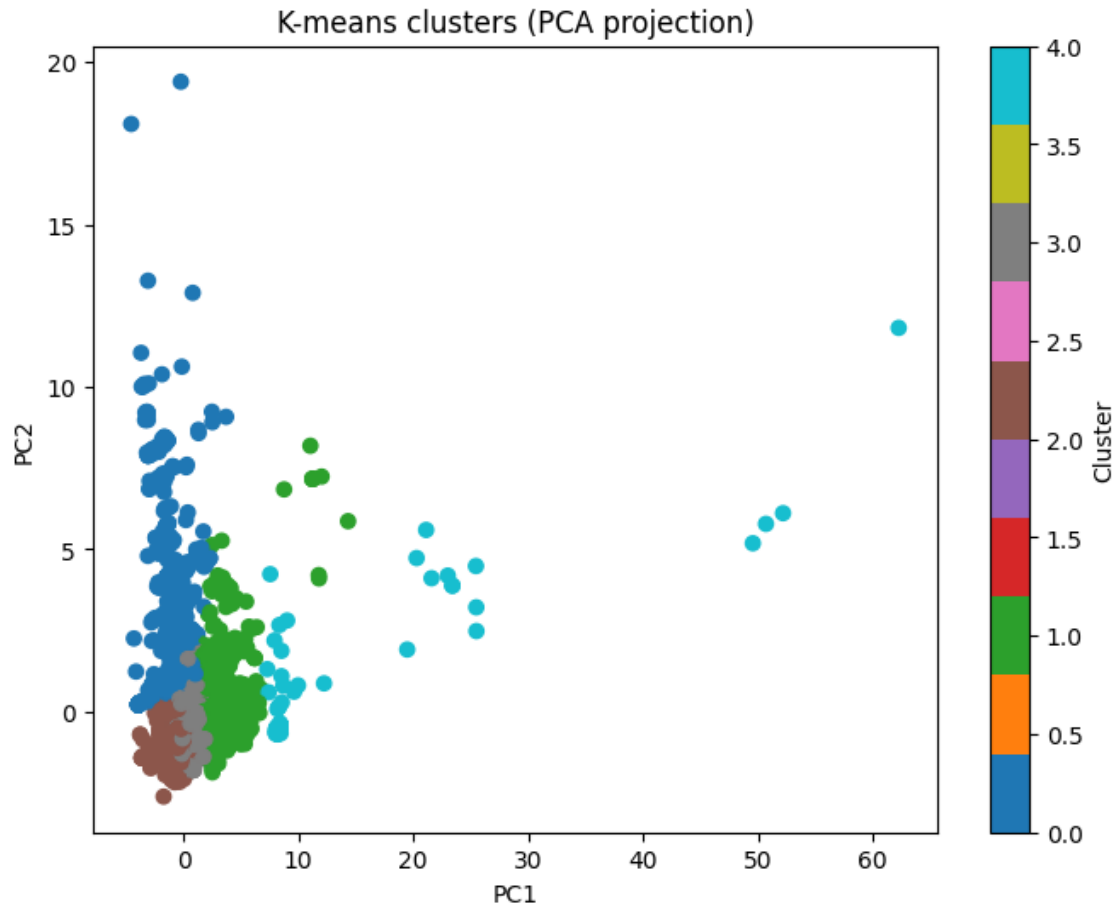


Fig 5. Scatterplot of PC1 and PC2 (PCA projection)

Observing the scatterplot, the more unusual combination of clusters like number 4 or 0 are a lot more scattered around and tend to fall on more extreme values, like it can be seen on both the scatterplot and the general cluster table.

As a test to measure the accuracy of the clustering a silhouette score was made to be able to quantify how close each point in the cluster was to each other. The silhouette score here was 0.3181. The closer it is to 1 the better, and the closer it is to -1 the worse.

The derived metrics and results from this cluster analysis for the amount of milk yield kg was the following:

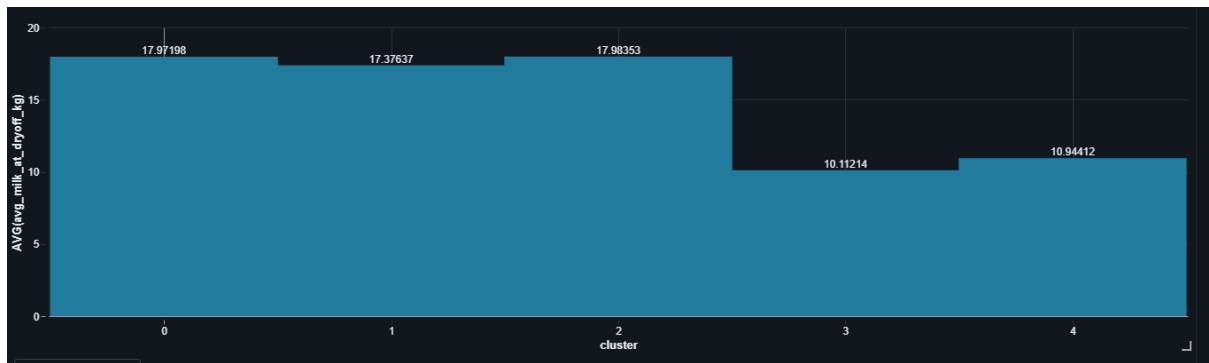


Fig.6. Bar chart showing milk yield kg at day of dry off per cluster

As it can be seen, the more populated clusters were to ones with the higher milk yield on average. This highlights the limitations of the strategies used for cluster 3 and 4, and points for improvements to be made for them.

6.2 Other methods for analyses

6.2.1 Using the new feature

Earlier on in the document I talked about doing a bit of feature engineering and making a new one for the cluster analysis called **has_pregnant_group**. The silhouette score achieved for this iteration was of: 0.4797. A slightly lower score, but by a very negligible amount.

The results of the extra feature and what numbers it had garnered might explain these results.

Cluster	has_pregnant_group (%)
0	0.039
1	0.057
2	0.074
3	4.907
4	0.028

Fig.7. Table indicating the avg usage of the has_pregnant_group rule in percentage value

After looking at the results, only a very small number of farms do end up using this strategy as it is not a very common one to use in situations. Aside from cluster number 4, which compared to the rest of the clusters has a very high amount of this happening. And this is helpful to determine the true strategies and methods for each cluster, as the 4th one has a clearer picture into what weights are used for the clustering itself.

6.2.2 Using ANN's

When I switched from traditional clustering to implementing an autoencoder into the cluster analysis, I went through many iterations of different versions of the analysis. I will subsequently talk about the most important ones. Because of the nature of neural network weight initialization and training, results can vary between runs even with the same hyperparameters. To deal with that, the model was trained and evaluated using 5 randomly chosen seeds, with each of the silhouette scores attached for each run. The mean and standard deviation of those 5 runs is calculated, rather than one individual score, making it a more statistically robust interpretation of model performance.

6.2.2.1 Base query, no extra feature and thresholding applied

In this iteration I used only the base query, which includes 4 features and no thresholding has been done. The number of clusters here is 5, running over 150 epochs and a bottle neck of 16.

The mean silhouette score ended up being 0.4436. The distribution of farms to clusters ended up being:

Cluster	Nr of farms
0	18706
1	4232
2	325
3	3
4	3251

Fig.8. Table indicating the distribution of farms per cluster

This distribution of the farms is quite unfavourable as having one of the clusters be populated by only 3 of the farms is not acceptable for this analysis. But even with that the silhouette score shows to be a big improvement to the regular clustering analysis. This might be saying something.

6.2.2.2 Extra feature and Thresholding applied

For this iteration I used the base query, with the new feature applied as well as the thresholding. The rest of the values stay the same, including the cluster number which stays at 5.

The mean silhouette score here is 0.3861. And the distribution is the following:

Cluster	Nr of farms
0	16131
1	658
2	1423
3	74
4	7653

Fig.9. Table indicating the distribution of farms per cluster

As observed, the silhouette score has gotten worse, but the distribution of the farms got a little bit better here. Still this distribution would still have to be classified as unfavourable as the distribution is still lacking for this analysis.

6.2.2.3 Only extra feature applied

In this iteration I only included the extra feature in the analysis. The rest was the same.

The silhouette score was 0.4859, and the distribution was the following:

Cluster	Nr of farms
0	7227
1	158

2	18615
3	456
4	26

Fig.10. Table indicating the distribution of farms per cluster

This time the score improved again, but the distribution was quite lacking again.

6.2.2.4 Only Thresholding applied

For this iteration, the only change that was made was the removing of the extra feature. Everything else stayed the same.

The mean silhouette score in this case was 0.3468. And the distribution ended up being:

Cluster	Nr of farms
0	15473
1	441
2	4158
3	4510
4	1361

Fig.11. Table indicating the distribution of farms per cluster

This time the results are a bit more interesting. The silhouette score for this iteration was lower than all the other scores that used ANNs, but as for the distribution this is the best we have seen so far. There is some kind of discrepancy here and a trend of something specific happening here.

6.2.2.5 Changing cluster size with no extra feature or thresholding applied

For the next iterations I will be choosing a cluster size of 4 to see how the distribution changes overall as well as to see how the values for each cluster change to the new cluster size. Like it was stated, this one has only 4 clusters and no extra feature and thresholding applied, everything else stays the same.

The mean silhouette score was 0.5110. And the distribution was:

Cluster	Nr of farms
0	328
1	5195
2	11
3	20989

Fig.12. Table indicating the distribution of farms per cluster

The silhouette score wasn't hindered at all, it actually improved as I reduced the cluster number, but the distribution once again, was not very good at all with far too few farms ending up in cluster 2.

6.2.2.6 Changing cluster size with extra feature and thresholding applied

For this next one the cluster number stays at 4, with the extra feature and thresholding applied to it.

The silhouette score for this iteration ended up being 0.5012, and the distribution of the farms was the following:

Cluster	Nr of farms
0	23570
1	655
2	1636
3	78

Fig.13. Table indicating the distribution of farms per cluster

Once again, the silhouette score here was very strong when the cluster number was reduced, achieving the second highest mean score in the whole analysis. But also repeating itself from the last iteration, the distribution continues to be quite poor. Both results having different distribution structures but both end up being similar when you weight the farms in each cluster comparatively.

6.2.2.7 Changing cluster size with only thresholding applied

In this iteration only the thresholding is applied, the rest continues being the same.

The silhouette score ended up being 0.3751, and the distribution was the following:

Cluster	Nr of farms
0	15488
1	1751
2	4190
3	4514

Fig.14. Table indicating the distribution of farms per cluster

Once again, after we removed the extra feature and only had the thresholding applied in the results, we see the same phenomenon happening again. The silhouette score dramatically decreased compared to the previous two iterations, but when observing at the distribution it looks to be a lot better distributed than the other two, making it look more suitable for analysis.

6.2.3 Discussion

After testing the autoencoder with K-Means model many times over, interchanging parameters throughout, the results I ended up getting weren't the ones I had hoped for exactly, but they were still quite interesting one's worth looking at.

There was a general pattern that emerged after running all these iterations. For this I will refer the main and different conditions at which the iterations were ran with as conditional metrics for the analysis. If you look at it closely this is what can be seen (for K=4):

Situation nr	Nothing applied	Extra feature applied	Thresholding applied	Result (silhouette score)
1	True	False	False	~0.51
2	False	True	True	~0.50
3	False	True	False	~0.50
4	False	False	True	~0.37

Fig.15. Iteration matrix testing different conditions

The key difference here between the different situations is that the last one has a lower average score than the rest of them, which they all seem to share around the same average score. The negative condition for these results is in the form of both the nothing applied and the extra feature applied. Whenever nothing applied is True the score always came out to ~0.51, and whenever the extra feature was True the score also always came out to ~0.50, regardless of if the thresholding was applied or not, it would always overpower it. Then as soon as the negative conditions were False and only Thresholding was True, the score always hovered at around ~0.37.

What this means is that the other two conditions, the negative conditions hold more weight when it comes to the process of calculating each cluster and its values. The reason for this is whenever the thresholding is applied it doesn't allow for any of the extreme values that I came across many times to be let through into the cluster analysis. The thresholding only takes into account the most normal values that can be found across all farms, and even if the thresholds are set for the absolute max values for each setting a farmer can put for their milk access settings, those values are still quite normal compared to the overblown ones that could be found before. But the thresholding re-organizes all the values but consequently makes separating each farm into different clusters harder since the values aren't so extreme here, which is the reason for the lower silhouette score.

The reason why the negative conditions take over, and why the scores tend to be higher is due to those extreme values making it into the analysis. It is much easier for the model to classify farms into each cluster when the extreme values are present, as those values almost always tend to be the overwhelming minority, which in a lot of cases ends up being a whole separate cluster or pair of clusters with those extreme values defining it. Therefore, it is easy to group together the extreme values compared to more normal ones, leading to a better score. That works the same for the extra feature condition. Since almost no farms happen to be using it, the cluster that populates it tends to be a huge minority. And in the results, the case tends to be that only one of the clusters is using said feature, leading to a better silhouette score. This is also supported with the scatter plot (PCA projection) made with only the thresholding condition applied:

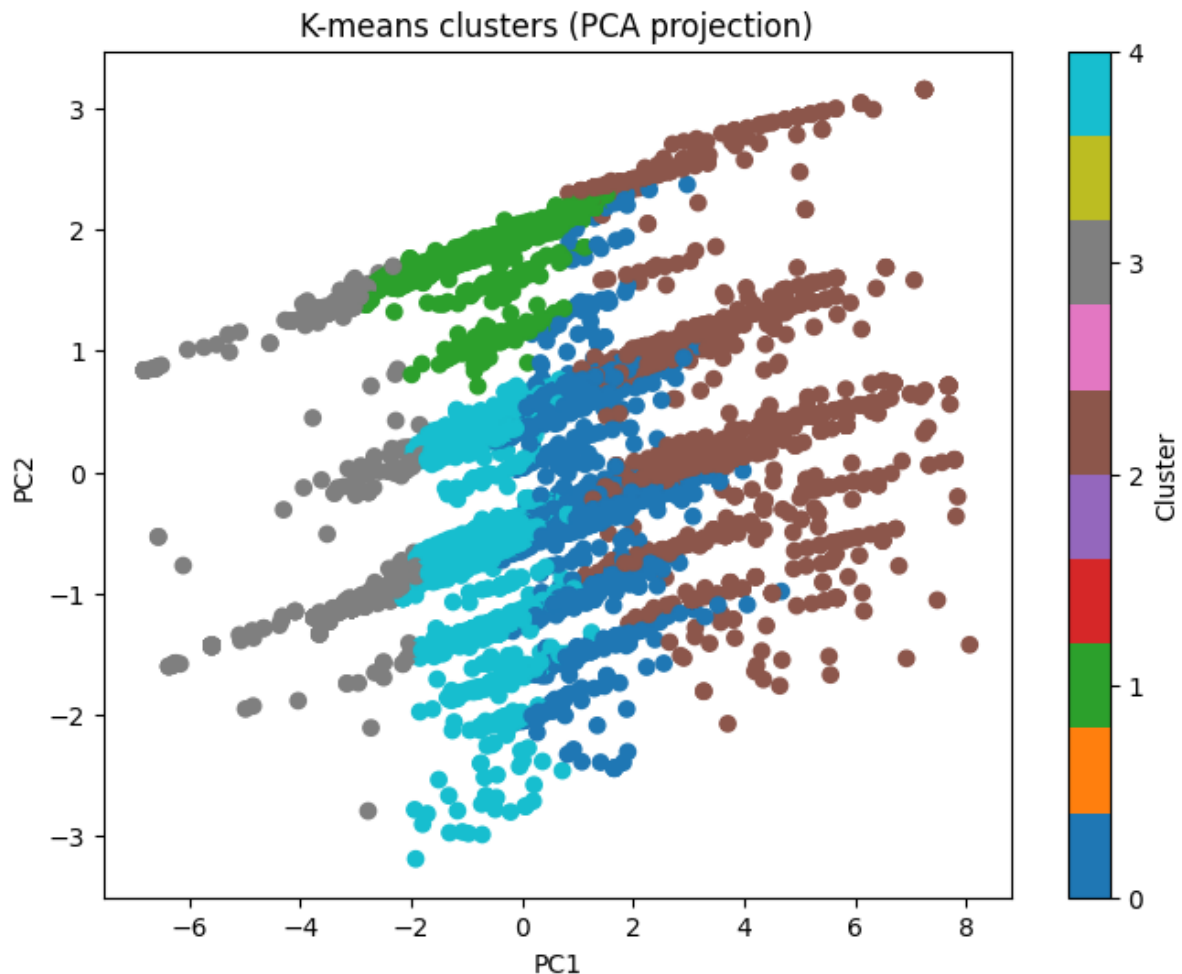


Fig.16. Scatterplot of PC1 and PC2 (PCA projection) with thresholding applied this time

The clusters are broadly separating along PC1 (horizontal axis), blue is mostly on the left, green on the right, grey in the middle top. This suggests the clustering has found some real structure. The diagonal stripe pattern the data forms diagonal bands rather than round blobs, which suggests the features have strong linear correlations with each other. This is common in MAS settings, farms that allow more milkings also tend to set higher yield thresholds etc. PCA is picking up on this linear structure.

The main differentiator of both graphs, like I have been explaining with the negative conditions, is 1 or 2 of the cluster being clear outliers and having extreme values. In the first scatter plot I featured, the extreme separation of the grey cluster is real information, it shows cluster 3 is different from everything else, not just slightly offset.

6.2.4 Interpretation

After seeing all of these results and going over the explanations to those results, it wouldn't be too farfetched to have the opinion that the new extra feature, the `has_pregnant_group` feature is obsolete in this analysis, not contributing a lot to it, and more-so hindering it in the long run. And that would be a fair assessment of the feature, but there is something underlying in these results that we can take back with us.

The fact that the values are so extreme is what hinders but also helps the analysis. After applying the extra feature condition, or not applying either of the conditions, what is revealed are the combination of settings that should not be used and helps distinguish between good strategies and the bad ones, depending on the farmers' goals, therefore those farms that are in the more extreme clusters have an easy path for improvement and know exactly what needs to be better.

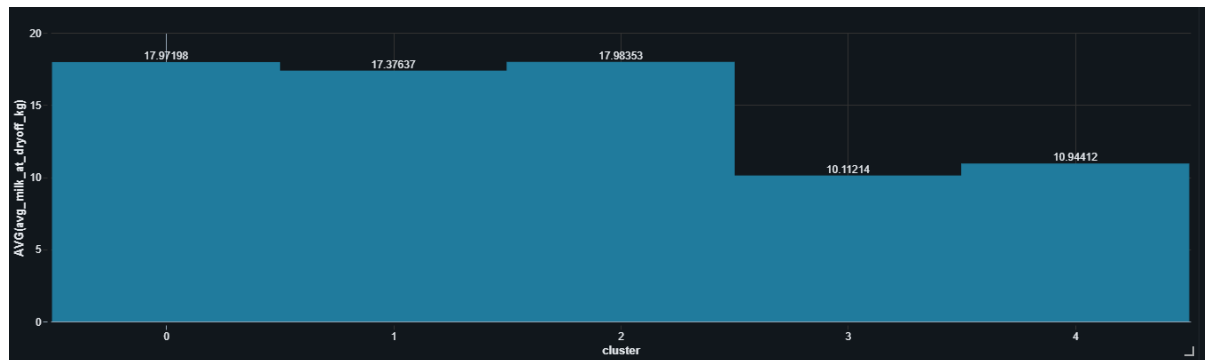


Fig.17. Bar chart showing milk yield kg at day of dry off per cluster

Going back to this graph, which shows the average milk yield at day of dry off for each cluster. All first three clusters yielding a high 18kg milk yield, while the last two showing only 10kg milk yield, which is a steep decline from the first three. This highlights the strategies that should be avoided and lets us focus on the farms that already currently have this kind of strategy in place. Which could lead to investigations on what is happening, why is the farmer setting it up like that, or if the farmer is even aware that they have those settings set the way they are.

It also puts the whole argument in an interesting position because there's two ways to look at it. The first is to look at it in a logical dry-off safety manner, in which the goal is to reduce the milk yield during the dry-off period, usually needing to get below the 15kg threshold for a healthy dry-off. The reason why milk yield is better to be lower during dry-off is if cows are yielding high amounts of milk at the end lactation stage it puts too much strain on the animals and then can become more susceptible to diseases such as mastitis and will yield less milk in the next stage of the cow. So in this case, the feature highlights what strategies farmers in theory should move closer to, to maintain a healthy dry-off of the animal. But it also highlights how to keep a high milk yield for farmers if that is what they desired. But in any case, this still raises concerns for what farmers are doing. This shows that farmers are still choosing to get the most out of their cows even at the dry-off stage, and even if it puts their own cows at risk.

7. Personal Reflection

Looking back at my whole internship period, from start to finish I think in my opinion it has been a very pleasant and worthwhile experience I went through during this time. In terms of personal growth as well as I got better in a technical sense at using my skills altogether.

During this internship I was exposed to SQL a lot, which previously I hadn't had that much experience with SQL aside from one other project I had. With this project here being so SQL heavy it really helped me improve my skills there by a wide margin compared to how I was at before. Things like my proficiency in data manipulation and working with/manipulating large, real life, structured datasets really improved. I also improved in my analytical skills, like my ability to identify patterns in complex datasets improved and all in all this internship gave me a better understanding of how data reflects real-world processes and how I would translate real-world problems into data questions.

For my personal development I also feel like I made some strides in that sense. Having finally experienced first hand how the professional and corporate worlds work was very insightful and useful to have seen and lived. Since it was my first time in a real professional environment, I had a lot of things to learn and for now this is only the start, but I would say that from this I have gotten a pretty good start and understanding of this new era that me and many of my classmates will now start for our careers.

Being able to work autonomously while also collaborating with teams of experts in the field from different departments was useful for my development. Working cross-functionally with co-workers of other teams and specialties, being able to translate my findings to a nontechnical person and have meaningful conversations with them that lead to good feedback and advice that I can take back with me for my analyses was an extremely useful experience that I regularly practiced with them. But also having total freedom over my project and over what I do is a point I can't forget to mention to bring up, as it gave weight to my decisions and made me solely responsible for my work. Working under that kind of pre-tense was another useful experience to take back with me. And the combination of having that autonomy over my work and having other colleagues to confide in, exchanging questions and solutions, is truly one of the more important skills I learned here.

7.1 Challenges and Limitations

The main challenges that I faced during this internship would have to be what I call, Domain Knowledge hump. There were a lot of very technical and niche farming terminology and general knowledge that I had never been exposed to before ever coming to this company. So a good portion of the start of this internship was about trying to understand the kind of data that I am working with and just doing some general exploration of the data so I can obtain that much needed context to be able to use this kind of data.

This was important because a good understanding on the data that one works with is crucial so that anything I was working with, any results that I would receive, any methods I would use, could continue having validity and above all else, it made sure I stayed in the loop of what I was working with and made sure I maintained a good grip on the concepts of my work.

A lot of this process of understanding the domain specific knowledge was about bridging together my skillset and my world with the world of this company that included such specific farming knowledge that no normal developer is going to be privy to, in the most usual cases.

And this mattered because using my skills properly in a way that made sense, and making sure I was not just completely wasting my time was a good point to keep in mind. I will mention that I talked about this to some of my colleagues and they said sometimes it takes years to understand this kind of data, which did startle me a little bit, but in the context of only my project it was not so bad and my understanding of the specific knowledge did not have to go so deep.

7.2 Closing statement

Overall, the whole experience left me with a good impression for the future. The co-workers I had were all very nice and welcoming. The team was very diverse in every sense, and there is good support from the company among other things.

For development goals I want to pursue in the future, I would like to explore and experience a different field of expertise within my domain. What exactly that could be I don't know yet, but I am always open to exploring different kinds of offers that I might have available to me. But in the same way, having the chance to improve upon the things I have learnt during this period would also be important to me. Even if I had a long exposure now to the types of things I am doing and the tools that I am using, there is always something else for me to learn, and that openness to learn new things, new skills, new methods, new tools and new domains is what makes me grow above my potential and is all I could ask for as a person who is looking to grow each year.

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